Princy Ranaivomanana

Ph.D. in Astrophysics | Data Scientist | Signal Processing

Antananarivo, Madagascar (open to remote / relocation)
rtprincy@gmail.com
rtprincy.github.io
github.com/rtprincy
linkedin.com/in/rtprincy
0000-0001-8470-0220



Professional Summary

Data scientist with a Ph.D. in Astrophysics and 5+ years of experience building machine-learning and signal-processing pipelines for large, noisy datasets. Specialised in unsupervised learning, anomaly detection, and time-series analysis. Strong knowledge of Python and Machine Learning libraries. Proven record of translating complex technical results for non-technical stakeholders.

Technical Skills

Programming	Python (NumPy, SciPy, Pandas, scikit-learn, TensorFlow, Keras, PyTorch), SQL, Bash, LaTeX
Machine Learning	Unsupervised clustering (UMAP, t-SNE), semi-supervised GANs, classification & anomaly detection on tabular, time-series, and image data
Time-Series & Signal Processing	Lomb–Scargle periodograms, Fourier analysis, periodicity detection in sparse / heteroscedastic data
Data Engineering	Automated QC pipelines, large-scale dataset processing, Git/GitHub, Docker, SLURM job scheduling
Communication	Cross-functional collaboration, stakeholder translation of technical results, technical writing, public speaking

Professional Experience

07/2026 – Present **Adjunct Supervisor for Master's Students, Astronomy**, University of Antananarivo, *Madagascar*

05/2026 – 08/2026 **AI Prompt Quality Assurance Specialist — Qwen**, Online

- Reviewed challenging physics problems/prompts (for Qwen 3.5-Plus) submitted by PhD-level contributors on the DataForce platform for the Eclipse project, and scored them against defined acceptance criteria.

02/2026 **Data Scientist — Cross-Industry ML Project**, Radboud University Modelling Camp

- Partnered with a multidisciplinary team to build a machine-learning model for genotype classification, developed for a nutrition-industry client.

10/2021 – 09/2025 **Doctoral Researcher**, Radboud University (Netherlands) & KU Leuven (Belgium)

- Built an unsupervised ML pipeline (UMAP, t-SNE) to detect 193 previously unknown patterns and filter contaminants from a noisy ∼60,000-record dataset, despite the absence of labelled training data.

- Designed and deployed Python signal-processing pipelines to extract periodic patterns from irregularly-sampled, heteroscedastic time-series data across four independent data sources.

2020 – 2021 **Master's Research Student — Generative Modelling**, University of Manchester, *United Kingdom*

- Developed semi-supervised GAN models for synthetic data generation and classification on large-scale survey data, improving pipeline performance for sparsely labelled datasets.

Education

2021 – 2025 **Ph.D. in Astrophysics**, Radboud University – KU Leuven, *The Netherlands / Belgium*

Thesis: machine learning & time-series analysis applied to large astronomical surveys

2020 – 2021 **M.Sc. in Astronomy**, University of Manchester, *United Kingdom*

Thesis: deep generative models for classification of survey data

2012 – 2017 **M.Sc. & B.Sc. in Astronomy / Physics**, University of Antananarivo, *Madagascar*

Selected Publications

8 peer-reviewed publications (5 as lead author) · full list: ORCID

1. Ranaivomanana, P. and Uzundag, M., "Exploring Late Stellar Evolution in the Era of Large Surveys: Machine Learning Prospects for Hot Subdwarfs and White Dwarfs," *Universe, 12, 257 (2026)*

2. Ranaivomanana, P., et al., "Unsupervised learning for variability detection with Gaia DR3 photometry," *A&A, 704, A70 (2025)*

3. Ranaivomanana, P., et al., "Variability in hot sub-luminous stars and binaries: machine-learning analysis of Gaia DR3 multi-epoch photometry," *A&A, 693, A268 (2025)*

Certifications

- Machine Learning in Production (*DeepLearning.AI*)
- Advanced Machine Learning and Signal Processing (*Coursera*)
- Generative AI: Foundation Models & Platforms (*Coursera*)
- Exploratory Data Analysis for Machine Learning (*Coursera*)
- Data Analysis with Python (*Coursera*)
- Data Fundamentals (*IBM SkillsBuild*)

Recognition & Leadership

- Invited speaker, Hamburg Observatory Colloquium (2025); 4 contributed talks and 5 poster presentations at international conferences (2022–2025).
- Recipient of 5 competitive research/travel grants, including 3 consecutive LKBF conference grants (2023–2025) and a Newton Fund/DARA scholarship (2020).
- Delivered data-analysis and instrumentation tutorials at Radboud University (2021–2024); supervised 4 undergraduate researchers at KU Leuven (2022).
- Active career mentor with CorrelAid (2026–present).

Last updated: October 29, 2026